

Boundary and initial conditions in EXHALE (EXHALE)

This memo documents the boundary conditions and the initial conditions of the EXHALE / EXHALE hydrodynamics solver at the level of the actual implementation: what is imposed, where in the code, why it has the form it does, and what dynamical behavior it produces (in particular the “breathing base”). File references are relative to the repository root; the structure is inherited from ATES v2 (Caldirola et al.) unless noted as an EXHALE extension.

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1. Overview: where and when the BCs act

1.1 Ghost-cell layout

The radial grid carries N computational cells ($j = 1, \dots, N$; $N = 500$ by default) plus $N_g = 2$ ghost cells on each side, so all state arrays are dimensioned $(1-N_g : N+N_g)$, i.e. $j = -1, 0$ below the base and $j = N+1, N+2$ above the top (`src/modules/init/parameters.f90`, `define_grid.f90`). The first interior cell $j = 1$ sits at the lower boundary radius $r = 1$ (lengths are normalized to the planetary radius R_0); the last interior cell $j = N$ sits at $r_{\max}$, which is either the Roche/Hill truncation $r_{\max} = (3M_*/M_p)^{-1/3} \tilde{a}$ in Roche mode or a user-chosen outer radius in spherical mode (`input_read.f90`).

All boundary conditions are applied to the *ghost cells only*; the interior cells $1, \dots, N$ are never overwritten. The fluxes at the two boundary interfaces then emerge from the same HLLC Riemann solver used everywhere else, with the ghost states as the outside input. In this sense every BC in the code is a *weak* (flux-mediated) condition: even the “Dirichlet” base values act on the interior only through the Riemann problem at the interface $j = 1/2$.

1.2 The three entry points (`Apply_BC.f90`)

All BC logic lives in one module, `src/modules/states/Apply_BC.f90`:

- `Apply_BC(u,u)` — conservative-variable wrapper: converts $U \rightarrow W$ (primitive), calls `Apply_BC_W`, converts back. Called after every Runge–Kutta stage in the main loop (`EXHALE_main.f90`), before every residual evaluation, and by the steady-state Newton solver each time it unpacks a trial state.
- `Apply_BC_W(W,W)` — the actual assignment of the four ghost states: lower ghosts via `BC_component_constrho` (§2.), upper ghosts via extrapolation (§3.).
- `Rec_BC(WL,WR,...)` — the same conditions applied to the *reconstructed* left/right interface states after PLM or WENO3 reconstruction (`Reconstruction.f90`), so that the boundary stencils of the higher-order scheme see BC-consistent data rather than raw extrapolated slopes.

In the Newton steady-state solver the ghost cells are deliberately *not* unknowns: the residual is defined on the interior state only, and `Apply_BC` regenerates the ghosts from the interior at every function evaluation (`src/modules/time_step/steady_newton.f90`). The BCs are therefore identical between the time-marching and Newton phases by construction.

1.3 Normalization constants tied to the base

The lower BC is intertwined with the code-wide normalization (`input_read.f90`):

$$\mu = \frac{1+4Y}{1+Y} m_H, \quad v_0 = \sqrt{\frac{k_B T_0}{m_H}}, \quad p_0 = n_0 m_H v_0^2 = n_0 k_B T_0, \quad b_0 = \frac{GM_p m_H}{k_B T_0 R_0}, \quad (1)$$

where $Y \equiv n_{\text{He}}/n_{\text{H}}$ is the He/H number ratio from `input.inp`, n_0 is the user-specified total *nuclei* number density at the base, and T_0 the base temperature. (The code variable `mu` is m_H itself; the composition factor appears separately as

rho_bc, see below.) Densities are stored as mass densities in units of $n_0 m_H$, number densities in units of n_0 , pressures in units of p_0 , temperatures in units of T_0 , velocities in units of the isothermal hydrogen sound speed v_0 .

The equation-of-state closure used to convert between the hydrodynamic pressure and the temperature handed to the ionization/radiation module is the dimensionless ideal-gas law including free electrons (composition.f90):

$$T = \frac{P}{n_{\text{tot}} + n_e}, \quad (2)$$

with n_{tot} the total *atomic* (nuclei) number density and n_e from calc_ne. Under the default eos_metals 1 policy (2026-06-13) both include the trace metals — their nuclei in n_{tot} and their electrons in n_e — consistently with the charge balance inside the ionization solver; eos_metals 0 in metals.inp restores the legacy H/He-only budget (§2.6).

2. Lower boundary: fixed reservoir with a one-way valve

The two lower ghost cells are set by BC_component_constrho (Apply_BC.f90, lines 32–54):

```
W(ghost,1) = rho_bc ! density: Dirichlet
if (valve_eps > 0) then ! velocity: one-way valve
  W(ghost,2) = 0.5*( W(1,2) + sqrt(W(1,2)**2 + valve_eps**2) )
else
  W(ghost,2) = max( W(1,2), 0.0 )
endif
W(ghost,3) = ntot_bc + dp_bc ! pressure: Dirichlet (dynamic)
```

Both ghost cells receive the *same* state (no gradient across them). The three components implement three distinct ideas, discussed in turn.

2.1 Density: a fixed-composition reservoir

The dimensionless ghost density is the constant

$$\rho_{\text{bc}} = \frac{1 + 4Y + \sum_X a_X A_X}{1 + Y}, \quad a_X \equiv \frac{n_X}{n_H}, \quad (3)$$

the mean gas mass per H/He nucleus in units of m_H (mass_per_H/(1+Y) in the code). Because mass densities are normalized by $n_0 m_H$, this pins the ghost cells to “H/He nuclei density = n_0 with the prescribed composition”: the hydrogen nuclei density at the base is $n_H = n_0/(1+Y)$, the helium density Y times that, and each metal $a_X n_H$. The base therefore behaves as an *infinite isochoric reservoir*: however much mass the wind removes, the boundary cell always refills to the same density.

The metal mass term $\sum_X a_X A_X \approx 0.016$ at solar abundances (a $\sim 1.2\%$ effect) is included by the default eos_metals 1 policy, which also adds the metal mass to calc_rho; eos_metals 0 restores the legacy H/He-only trace approximation, $\rho_{\text{bc}} = (1 + 4Y)/(1 + Y)$.

2.2 Pressure: dynamically updated to pin the base temperature

The ghost pressure is $p_{\text{gh}} = n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}$, where $n_{\text{tot},\text{bc}} = (1 + Y + \sum_X a_X)/(1 + Y)$ is the (constant) total nuclei density at the base in units of n_0 ($n_{\text{tot},\text{bc}} = 1$ for H/He only) and δp_{bc} starts at the placeholder 10^{-10} and is recomputed after every ionization-equilibrium solve as the base electron density (`ionization_equilibrium.f90`):

$$\delta p_{\text{bc}} = \frac{n_{\text{HII}} + n_{\text{HeII}} + 2n_{\text{HeIII}} + \sum_i q_i n_i^{\text{met}}}{n_0} \Big|_{j=1-N_g} = \frac{n_e(\text{base})}{n_0}, \quad (4)$$

with q_i the metal ion charges (included under `eos_metals 1`, dropped under `eos_metals 0`). The point of this correction is visible through the EOS closure (2): at the ghost cells $n_{\text{tot}} = n_{\text{tot},\text{bc}}$ (by the density BC) and $n_e = \delta p_{\text{bc}}$, so

$$T_{\text{gh}} = \frac{n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}}{n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}} = 1 \quad \Longleftrightarrow \quad T = T_0. \quad (5)$$

In other words, the pair (fixed ρ , dynamically corrected p) is really the pair (*fixed nuclei density* n_0 , *fixed temperature* T_0): as photoionization develops at the base and free electrons add partial pressure, the ghost pressure is raised by exactly the electron share so that the base stays isothermal at T_0 instead of being artificially cooled. For a near-neutral base (the usual situation, since the base optical depth is huge) δp_{bc} is tiny and the BC reduces to $p = p_0$.

The metal-electron share of δp_{bc} is small (even with all low-ionization-potential metals singly ionized at the base it is $\lesssim 4 \times 10^{-4}$), but it is carried anyway under `eos_metals 1` for consistency with `calc_ne`.

2.3 Velocity: the one-way valve

The ghost velocity copies the first interior cell when the gas moves outward and is clamped at zero when it moves inward:

$$v_{\text{gh}} = \max(v_1, 0). \quad (6)$$

Neither a wall ($v_{\text{gh}} = 0$) nor a free condition ($v_{\text{gh}} = v_1$) would be physical here:

- a *wall* would starve the wind — the escaping flow is launched by photoionization heating *above* the base, and the mass it carries must be resupplied through the lower boundary;
- a *free* condition combined with the fixed-density ghosts would let the reservoir act as an unphysical mass sink whenever the interior gas sags downward, swallowing arbitrary amounts of gas at fixed ρ_{bc} .

The valve resolves the asymmetry: *outflow is allowed (the reservoir feeds the wind), sustained inflow is refused (the reservoir does not absorb a backflow)*. Physically the boundary models the top of an essentially infinite, hydrostatic lower atmosphere held at (n_0, T_0) .

Smooth valve (EXHALE extension). The hard max has a kink at $v_1 = 0$ — and, as discussed in §2.4, $v_1 = 0$ is precisely where the base dynamics lives. The kink is invisible to the explicit time-marching scheme but fatal to the steady-state Newton solver, whose line search cannot cross a non-differentiable point of the residual. The input key `Valve eps`: therefore enables the smooth approximation

$$v_{\text{gh}} = \frac{1}{2} \left(v_1 + \sqrt{v_1^2 + \varepsilon^2} \right), \quad (7)$$

which is C^∞ , tends to v_1 for $v_1 \gg \varepsilon$, tends to 0 for $v_1 \ll -\varepsilon$, and biases the valve by at most $+\varepsilon/2$ near $v_1 \sim 0$. With $\varepsilon = 0$ (default) the legacy hard valve is used.

2.4 The “breathing base”

Because the base velocity is a free variable governed by an *asymmetric* boundary (an open pipe outward, a wall inward), the near-base flow does not settle onto the exact $v = 0$ fixed point but onto a small limit cycle around it. One cycle proceeds as follows:

1. **Exhale** ($v_1 > 0$): the wind draws mass up; the Riemann problem against the fixed- ρ_{bc} ghosts supplies gas at base density — the boundary acts as a mass source.
2. **Overshoot**: if momentarily more mass is lifted than the heating profile above can carry, the over-supplied column begins to sink back under gravity; v_1 turns negative.
3. **Inhale** ($v_1 < 0$): the valve closes ($v_{gh} = 0$). The descending gas piles up on the closed boundary, raising ρ and p in the first few interior cells — a transient wall reflection.
4. **Rebound**: the accumulated pressure pushes the column back out; v_1 turns positive and the cycle repeats.

This inhale/exhale oscillation is the *breathing base*. It is an intrinsic consequence of the valve-type BC — present in the original ATES as well — and not a numerical defect. Two regimes should be distinguished (see docs/initial_condition_benchmark.pdf, §7):

Self-limiting (normal). In the time-marching hydrodynamics the pressure built up during the inhale phase immediately produces the restoring force of step 4, so the amplitude saturates. The oscillation stays confined to the first few cells ($r \lesssim 1.1 R_p$) and has no measurable effect on the wind region or on $\dot{M}$.

Self-amplifying (pathological). If the *global* initial condition is inconsistent (the canonical example: a naive full-Parker density profile, whose hot column drains onto the base), each cycle deposits more mass than the previous one and the breathing grows until the run diverges. The cure is a consistent IC (hydrostatic density, optionally with a warm seed overlay), not a change of the BC.

Numerical consequences. Harmless as it is physically, the breathing base interacts with three parts of the code, each of which carries a dedicated counter-measure:

- *Convergence monitoring.* The breathing cells never stop oscillating, so a convergence criterion of the form $\max_j |\Delta u|$ over the whole grid can stall at the 10^{-2} – 10^{-1} level indefinitely. ATES therefore evaluates its momentum-flatness and du/dt diagnostics only over the wind region $[j_{\min} : N]$, where $j_{\min}$ is the first cell with $r \geq r_{\text{esc}}$ (the Escape radius input; §4.2). Cases with unusually violent breathing (HD 189733 b is the standing example) can still fail the strict du threshold; reducing CFL, modifying the valve, or replacing the boundary Riemann solver did *not* remove the oscillation in our experiments — it is robustly attached to the BC type. A periodic Shapiro (1970) low-pass filter, the stabilizer the reference code CETIMB (Koskinen et al. 2013a) applies to exactly this gravity-unbalanced sound-wave instability, *does* damp the oscillatory component (§2.5), but it biases the momentum and pushes the clean cold-IC cases off the wind branch into infall, so it is kept opt-in rather than a default.

- *Newton solver.* The $v_1 = 0$ kink of the hard valve sits exactly on the breathing orbit; the smooth valve of §2.3 exists for this reason.
- *Advection post-processing.* The `_adv` correction integrates ionization states along streamlines; during the inhale phase ($v \leq 0$) the streamline of a base cell exits the domain through the lower boundary and the correction is ill-defined (this produced the historical temperature spike/sawtooth at $r \approx 1.1$). The post-processor therefore keeps the equilibrium solution and skips the advection correction wherever $v \leq 0$ at the base.

2.5 Optional stabilizers and the selectable base mode (EXHALE)

EXHALE adds three opt-in lower-boundary controls, ported while studying the breathing/infall behavior against CETIMB (Koskinen et al. 2013a; Huang et al. 2023). All three are *off by default*, so the legacy behavior is unchanged unless requested. Their measured effects are tabulated in §7..

Shapiro filter (Shapiro filter: `<eps> [<every>]`). Every every steps a weak two-step Shapiro (1970) 1–2–1 low-pass filter $u_j \leftarrow \frac{\epsilon}{4}(u_{j-1} - 2u_j + u_{j+1})$ is applied to the conservative variables. This is CETIMB’s cure for the gravity-unbalanced sound waves and it does smooth the breathing transient. *However*, on the otherwise-clean cold-IC cases it injects a systematic momentum bias that drives the solution off the transonic-wind saddle into the infall branch ($v < 0$ almost everywhere; see the `_Shap` rows of Table 2). It is therefore opt-in — useful only for cases that genuinely breathe, not as a global default.

Mass-flux base velocity (Base velocity: `massflux`). Instead of the one-way valve (§2.3), the base velocity is set from the steady mass-flux continuity $v_0 = F_c / (\rho_0 r_0^2)$ with F_c a slow exponential moving average of $\rho v r^2$ over the wind region. It is physically benign — it never causes infall on its own and gives the cleanest base (it suppresses the $\pm \text{m s}^{-1}$ valve oscillation) — but it slows convergence ($\sim 5 \times$ more steps), so the fast valve stays the default.

Pressure-anchored base (Base BC: `pressure [<p_ubar>]`). Instead of fixing the base *number density* at n_0 (§2.), this fixes the base *pressure* at `p_ubar` microbar (default $1 \mu\text{bar}$, the CETIMB lower boundary) and derives n_0 from $p = n_0 k_B T_0 n_{\text{tot, bc}}$. Because the isothermal base is over-determined (it already pins $T = T_0$), this is a re-parameterization of the same ghost; its physical effect is simply a much less dense base ($1 \mu\text{bar} \Rightarrow n_0 \approx 5 \times 10^{12}$, $\sim 20 \times$ below 10^{14}), which shrinks the dense-base gravity source ρg that feeds the breathing. Empirically a density-anchored run at the same low n_0 ($\log n_0 = 12.5$) gives an almost identical converged wind, confirming that the lever is the base *density*, not the anchoring variable (Table 2).

Implementation note. The per-cell ionization-equilibrium sweep is now OpenMP-parallel over cells (the metal- and charge-exchange per-cell coefficients are kept thread-private); the parallel result is bit-identical to the serial one, so none of the BC/IC behavior below is affected. See `docs/openmp_parallelization.md`.

2.6 Temperature, ionization, and metals at the lower boundary

No boundary condition is imposed on the temperature or on any ionization fraction. The ghost cells are included as ordinary cells in the ionization-equilibrium sweep (`ionization_equilibrium.f90` loops over $j = N + N_g, \dots, 1 - N_g$), so

their T , H/He stages, He 2^3S , and all metal stages come out of the same coupled MINPACK solve as the interior, evaluated at the ghost state $(\rho_{bc}, n_{\text{tot},bc} + \delta p_{bc})$. Combined with §2.2 this makes the effective thermodynamic boundary condition “ $n = n_0$, $T = T_0$, ionization in local equilibrium”. Metal abundances enter only through the runtime `metals.inp` fractions; the boundary needs no special metal treatment: their abundances ride on the (fixed-composition) density BC, and under the default `eos_metals 1` policy their mass, electrons, and nuclei are carried consistently through `calc_rho/calc_ne/calc_ntot` (`eos_metals 0` restores the legacy trace approximation in which they never appear in the mass or momentum budget).

3. Upper boundary: supersonic free outflow

The upper ghost cells are filled by extrapolation (`Apply_BC_W`, lines 73–77):

```
do k = 1, Ng
  W(N+k,:) = W(N,:) ! PLM / first order
  if (use_weno3) W(N+k,:) = 2*W(N+k-1,:) - W(N+k-2,:) ! WENO3
enddo
```

i.e. zeroth-order (copy) extrapolation of all primitive variables for the PLM scheme, upgraded to linear extrapolation when WENO3 reconstruction is active (the wider stencil of WENO3 is sensitive to the artificial flattening a plain copy would imply). `Rec_BC` applies the same rule to the reconstructed interface states.

The justification is characteristic-based: by the time the flow reaches r_{max} the wind is expected to be supersonic, so all three characteristics leave the domain and no physical information may be prescribed from outside; extrapolation is then the consistent “transparent” closure. Two caveats follow directly:

- If the flow is *not* supersonic at r_{max} (observed for deep Roche-lobe-overflow geometries, e.g. WASP-121b Case D, which is subsonic at L1), one in-going characteristic formally remains and the extrapolation BC is no longer strictly well-posed; in practice the scheme remains stable but the outer solution should be treated with caution (geometry-limited regime).
- The BC carries no information about the medium beyond r_{max} (stellar wind, confinement pressure); EXHALE models the planetary outflow only.

No ionization or temperature condition is applied at the top either; the ghost cells participate in the equilibrium sweep like all others, with their extrapolated (ρ, p) .

4. Boundary-adjacent settings

These are not boundary conditions of the hydrodynamic update, but they are boundary choices of the model and are easily confused with BCs.

4.1 Radiative transfer: irradiation from the top

The stellar EUV/X-ray flux enters through the *outer* boundary. The column densities that attenuate the ionizing flux are integrated top-down (`calc_column_dens`, `utilities.f90`): the outermost ghost cell is assigned its own local column $N(N+N_g) = n \Delta r$, and the recursion $N(j) = N(j+1) + n_j \Delta r_j$ proceeds inward, i.e. the optical depth is (essentially) zero at the top of the grid and grows toward the base. This is the radiative-transfer analogue of a boundary condition: the unattenuated stellar spectrum is imposed at $r_{\max}$, and the base receives whatever survives the overlying column. Because the base column is huge, the lower boundary is effectively in radiative shadow — consistent with holding it at (n_0, T_0) .

4.2 Escape radius and the convergence window

The `Escape radius [R_p]`: input (r_{esc} , default 1.5) defines $j_{\min}$, the first cell with $r \geq r_{\text{esc}}$ (`define_grid.f90`). All steady-state diagnostics — momentum-flux flatness $\rho v r^2 = \text{const}$, du , dtu — are evaluated over $[j_{\min} : N]$ only, deliberately excluding the breathing base (§2.4). A guard added for deep-RLOF geometries (EXHALE): if $r_{\text{esc}} \geq r_{\max}$ the window would be empty (and `maxval/minval` over an empty section return $\mp \text{HUGE}$, producing a spurious instant “convergence”); in that case the window is clamped to the outer half of the domain and a loud warning asks the user to lower r_{esc} .

4.3 Domain truncation (where the upper boundary sits)

In Roche mode the grid is truncated at the Hill radius $r_{\max} = (3M_*/M_p)^{-1/3} \tilde{a} \approx L1$; in spherical mode at the user-supplied `Outer radius [R_p]`. The choice changes where the free-outflow BC is applied, and hence how defensible the supersonic assumption is (§3.); it does not change the BC itself.

5. Initial conditions (`set_IC.f90`)

Three concepts to fix first: cold IC, warm IC, and starter. For the reader’s convenience, every IC family below is a variation on three ideas. The distinction matters because it controls whether the run starts *inside* the wind basin or has to be pushed there.

Cold IC. The default. A fully *neutral*, isothermal ($T = T_0$) *hydrostatic* atmosphere with only a small outward velocity seed. Nothing is pre-heated or pre-ionized: the simulation grows the temperature, ionization, and wind from scratch. Because it is built from EXHALE’s own base BC (fixed ρ_{bc}, T_0), it is automatically *consistent with the lower boundary* and relaxes cleanly to the transonic wind.

Warm IC (warm seed / warm start). An initial state that already carries a *heated, ionized, flowing* profile, overlaid on the hydrostatic density — the transonic isothermal-wind and hot-Parker families. A plausible wind therefore exists from step 0, which is needed where a cold start cannot launch (deep-RLOF or low-gravity boil-off). It is still EXHALE’s own base with a thermodynamic overlay, so it stays consistent with the lower boundary; it only auto-enables `force_start` so the heating can settle.

Starter (starter solution / seed). A complete, externally pre-converged steady wind solution read in wholesale as the IC — the Wind-AE warm start (IC mode: `windae`), which ramps a Wind-AE `windsoIn` (or a grid/database seed) to this planet. Unlike a warm seed, a starter *replaces the entire state* with another code’s solution, including its own base; that base can clash with EXHALE’s lower BC and pull the run off the wind branch (§7.). A starter bootstraps hard cases but is not guaranteed consistent with EXHALE’s boundary.

When a run does not restart from saved profiles (§5.5), the initial state is built by `set_IC` in `src/modules/init/`. Three IC families exist; the choice is made by optional `input.inp` keys, with the cold hydrostatic atmosphere as the default:

Family	Key	Default
Cold hydrostatic + velocity seed	(none)	yes
Transonic isothermal wind	Transonic IC: <code>True</code>	off
Hot-Parker warm seed	Hot Parker IC: T_{wind} [K]	off
Automatic selection	IC mode: <code>auto</code>	off
Wind-AE warm start	IC mode: <code>windae</code>	off

The two non-default families force the first 1000 iterations (`force_start`) so that the heating structure can develop before the convergence test is allowed to fire. Whatever the family, the full dimensional IC is dumped to `output/IC_dump.txt` for inspection before the first step.

Automatic selection (IC mode: `auto`). `select_IC_auto` (`set_IC.f90`) picks the family from a single exact probe — `find_sonic` at the *cold* base sound speed in the actual potential. An interior sonic point selects the transonic IC; this catches both deep-RLOF bases (the Roche $\phi' \rightarrow 0$ topology toward L1 always crosses the critical condition; e.g. WASP-121b Case D, $r_c = 1.30$) and low-gravity boil-off planets ($r_c \simeq b_0/2c_0^2$ falls inside the domain for small b_0). Otherwise the cold hydrostatic default is kept (classic EUV-heated winds: HD 209458 b has $r_c \sim 54 R_p$, far outside the domain). There is no tunable threshold; b_0 (= the escape parameter Λ) is logged as a regime diagnostic, the warm seed is never auto-selected (the benchmark showed it buys nothing when the cold start works), and the explicit keys override `auto`. Design rationale and validation: `docs/auto_ic_design.md`.

Wind-AE warm start (IC mode: `windae`). With this key, `init.f90` builds the initial condition by running the in-tree Fortran port of the Murray-Clay/Broome *Wind-AE* steady-state wind solver (`src/modules/wind_ae/`). The bridge `wae_exhale_bridge` maps the planet’s EXHALE parameters ($M_p, R_0, M_*, a, F_{\text{XUV}} = J_{\text{XUV}}, L_*, \text{He/H, domain mode}$) to Wind-AE, ramps a shipped seed solution (`inputdata/windae_seed.csv`) to the planet by continuation, runs the BVP relaxation + outward integration, and writes `output/{Hydro_ioniz, Ion_species}_IC.txt` on the EXHALE grid; `load_IC` then ingests them. The whole EXHALE-input $\rightarrow$ Wind-AE solve $\rightarrow$ IC $\rightarrow$ run is a single EXHALE.x invocation. This is a metal-free H/He *warm start* (EXHALE re-equilibrates the ionization and any metals from the first step); finish with `Solver: Newton` for a quantitative $\dot{M}$. Best suited to seed-adjacent hot Jupiters; far-from-seed planets may fail the ramp (the run then advises IC mode: `auto`). A standalone generator `wind_ae_ic.x` (`make wind_ae_ic`) does the same out of process. Full port/validation: `md_backup/wind_ae_fortran_port_plan.md`.

The guiding principle in all three cases is *consistency with the lower boundary condition* of §2.: every IC matches $(\rho, p, v) = (\rho_{bc}, n_{tot, bc} + \delta p_{bc}, 0)$ at the base exactly, so the boundary never has to fight the initial state. The quantitative comparison of the families is in docs/initial_condition_benchmark.pdf; the conclusions quoted below come from there.

5.1 Default: cold neutral hydrostatic atmosphere + velocity seed

The original ATES initial condition, kept as the EXHALE default because it is the most robust (§5.3).

Density. An isothermal hydrostatic atmosphere in the (Roche or spherical) potential, anchored to the base density BC:

$$\rho(r) = \rho_{bc} \exp \left[b_{0, \text{eff}} (\phi(r_0) - \phi(r)) \right], \quad (8)$$

with one deliberate distortion: $b_{0, \text{eff}}$ starts at the physical value (1 in code units, the potential already carrying the Jeans parameter b_0) and is increased in steps of 0.2 until the density at the domain midpoint $r_{1/2} = (1 + r_{\text{max}})/2$ falls below 10^{-8} (set_IC.f90, lines 85–101). For low-gravity planets the true isothermal atmosphere would not decay enough across the domain; the loop artificially *steepens* the initial atmosphere so that the outer half of the grid starts effectively empty. The profile is then floored at $\rho = 10^{-8}$ (line 112). Equation (8) with $b_{0, \text{eff}} > 1$ is therefore *not* an equilibrium — it is a concentrated cold reservoir that the subsequent heating will evaporate.

Velocity. A small outward linear seed,

$$v(r) = \frac{1}{2} (r - r_0), \quad (9)$$

zero at the base and growing outward (line 108). Starting from exact rest delays the launch of the wind; the seed gives the outflow a direction without imposing any particular magnitude.

Pressure, temperature, ionization. Isothermal at the base temperature and consistent with the lower BC (lines 137–146):

$$p = (n_{tot, bc} + \delta p_{bc}) \frac{\rho}{\rho_{bc}}, \quad T = 1 (= T_0 \text{ everywhere}), \quad (10)$$

and a mostly neutral composition: $x_{\text{HII}} = \delta p_{bc} \sim 10^{-10}$, $x_{\text{HeII}} = 10^{-10} Y$, HeIII and He 2³S zero, the rest neutral. At the base cell this reproduces $(\rho_{bc}, n_{tot, bc} + \delta p_{bc}, T_0)$ exactly.

The physical picture: *a cold, neutral, hydrostatically stratified atmosphere is exposed to the stellar EUV flux at $t = 0$ and left to evaporate*. The first ionization-equilibrium sweep deposits the heating, the upper layers warm and expand, and the wind launches self-consistently — typically within the forced-start phase.

5.2 Transonic isothermal-wind IC (EXHALE)

Motivation: deep Roche-lobe-overflow geometries (the canonical case is WASP-121b Case D, whose sonic point sits essentially at L1) launch *no* wind from the hydrostatic IC — the run stalls in a bound, breathing state. For these cases

the run must start from a flowing solution. With `Transonic IC: True`, `wind_profile` (`set_IC.f90`, lines 177–233) solves the steady isothermal wind in the actual ATES potential via its Bernoulli integral:

$$\frac{1}{2}v^2 - c^2 \ln v - 2c^2 \ln r + \phi(r) = B, \quad c^2 = \frac{n_{\text{tot,bc}} + \delta p_{\text{bc}}}{\rho_{\text{bc}}}, \quad (11)$$

where c is the dimensionless isothermal sound speed of the cold base gas. The constant B is fixed at the sonic point r_c , located by bisection on the critical condition

$$\phi'(r_c) = \frac{2c^2}{r_c} \quad (12)$$

(`find_sonic`; in Roche mode $\phi' \rightarrow 0$ toward L1, so a sign change is generic). At each radius Eq. (11) has one subsonic and one supersonic root; the transonic wind takes the subsonic branch for $r < r_c$ and the supersonic branch for $r > r_c$ (robust bisection per cell, `wind_root`). The density then follows from steady mass conservation pinned to the base BC,

$$\rho v r^2 = \rho_{\text{bc}} v(r_0) r_0^2. \quad (13)$$

If no interior sonic point exists (gravity dominates out to r_{max} : a bound “breeze” case), the routine reports failure and `set_IC` falls back to the hydrostatic IC of §5.1 with a console notice. If the base itself is already super-critical, the flow is taken supersonic from the base.

5.3 Hot-Parker warm-seed IC (EXHALE)

This family is the product of the initial-condition benchmark (`docs/initial_condition_benchmark.pdf`, §7), whose central negative result deserves restating: **seeding the full isothermal Parker wind — its density *and* velocity — makes the run diverge**. The hot, low-density Parker column is globally inconsistent with the non-isothermal ATES equilibrium; it drains onto the base and drives the self-*amplifying* breathing mode of §2.4 (whereas the cold hydrostatic IC breathes self-limitingly and converges). The lesson: the base may not be handed an initial state whose global column it cannot support.

The robust compromise, enabled by `Hot Parker IC: Twind` (default $T_{\text{wind}} = 10^4$ K), is a *warm seed*: keep the stable cold hydrostatic *density* of §5.1 and overlay only the fields whose relaxation is slow and stiff — temperature, ionization, and a velocity head-start. The overlay is anchored to the cold base by a spatial smoothstep window (lines 114–133),

$$s(r) = \text{clip}\left(\frac{r-1}{r_{\text{tr}}-1}, 0, 1\right), \quad \xi(r) = s^2(3-2s), \quad (14)$$

with $r_{\text{tr}} = \text{hp_base_rtr} = 1.3 R_p$ (a compile-time parameter in `parameters.f90`), so that $\xi = 0$ at the base and $\xi = 1$ for $r \geq r_{\text{tr}}$. The overlaid fields are

$$\begin{aligned} T(r) &= 1 + \left(\frac{T_{\text{wind}}}{T_0} - 1\right) \xi, & v(r) &= v_p(r) s, \\ p(r) &= \rho \frac{n_{\text{tot,bc}} + \delta p_{\text{bc}}}{\rho_{\text{bc}}} T(1 + \xi), & x_{\text{HII}}, x_{\text{HeII}} &: \text{neutral} \rightarrow \text{ionized as } \xi, \end{aligned} \quad (15)$$

where the $(1 + \xi)$ factor in the pressure is the electron partial pressure of the ionized warm gas (particle number doubles at full ionization), and v_p is the Parker velocity head-start: the transonic profile of §5.2 re-solved with the *warm, ionized* sound speed

$$c_{\text{warm}}^2 = \frac{n_{\text{tot,bc}} + \delta p_{\text{bc}}}{\rho_{\text{bc}}} \frac{T_{\text{wind}}}{T_0} \times 2, \quad (16)$$

its *density discarded* (that is the part that destabilizes), and its velocity multiplied by s so it vanishes at the base. If the warm profile has no interior sonic point, the head-start quietly degrades to the small linear seed.

At the base $\xi = 0$ gives $T = T_0$, $p = n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}$, $v = 0$ — the lower BC of §2. exactly, with no inverted gradients for the first step to undo.

Benchmark verdict. The warm seed converges to the same $\dot{M}$ as the cold IC with no NaN/blow-up, but yields *no speedup* (−2.3% steps in the benchmark): ATES convergence is gated by the slow momentum (du) relaxation, not by the thermal/ionization relaxation the warm seed shortcuts. The cold hydrostatic IC therefore remains the default; the warm seed is kept for cases where the cold start struggles to launch.

5.4 Metals and common finalization

In every family the metals start maximally neutral: the entire elemental abundance is placed in the neutral ground stage,

$$f_{X^0} = \frac{n_X/n_{\text{H}}}{1+4Y}, \quad f_{X^+} = f_{X^{++}} = 0 \quad (17)$$

(set_IC.f90, lines 152–162; abundances from metals.inp). The first ionization-equilibrium sweep redistributes each element over its stages immediately, so the choice costs at most one equilibrium solve. The complete IC (dimensional r, n, v, p, T) is written to output/IC_dump.txt before the first step.

5.5 Restarts (Load IC)

When Load IC is enabled, set_IC is bypassed and the previous run’s profiles (including ghost cells) are read back from the *_IC.txt copies (load_IC.f90). This only sets the initial state; Apply_BC overwrites the ghost cells from the restored interior at the first step, so a restart cannot smuggle in a different boundary condition. (Operationally: the output/ → *_IC.txt copy is performed by the GUI, not by the Fortran code — for scripted runs the *_IC.txt files must be seeded by hand, or the code stops on the empty files it creates.)

6. Summary

The lower boundary is best read as a single physical statement — *the wind is rooted in an infinite isothermal reservoir at (n_0, T_0) that supplies but never absorbs mass* — implemented as fixed ρ , electron-corrected fixed p , and a one-way velocity valve. Its characteristic dynamical signature is the self-limiting breathing oscillation of §2.4, which is excluded from convergence diagnostics via r_{esc} , smoothed for the Newton solver via Valve_eps, and guarded against in the advection post-processor. The upper boundary is a plain supersonic free-outflow extrapolation whose validity rests on the wind crossing the sonic point inside the domain.

The initial conditions are built to respect the same reservoir: all three families (§5.) match the base state ($\rho_{\text{bc}}, n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}, v=0$) exactly, and the benchmark experience condenses into one rule — *the density column must be one the base can support*. The cold hydrostatic atmosphere (default) satisfies this trivially and evaporates into the wind on its own; the transonic IC replaces it with a flowing solution where the hydrostatic start cannot launch (deep-RLOF, sonic point at L1); the hot-Parker warm seed may overlay temperature, ionization, and velocity on the

Quantity	Lower boundary (base, $r = 1$)	Upper boundary ($r = r_{\max}$)
Density ρ	Dirichlet: $\rho_{bc} = \frac{1+4Y}{1+Y}$ (nuclei = n_0)	copy / linear extrapolation
Velocity v	one-way valve $\max(v_1, 0)$ (smooth opt.)	copy / linear extrapolation
Pressure p	$n_{\text{tot},bc} + \delta p_{bc}$, $\delta p_{bc} = n_e(\text{base})/n_0$	copy / linear extrapolation
Temperature T	implied: $T = T_0$ exactly (§2.2)	implied by extrapolated (ρ, p)
Ionization, metals	equilibrium solve in ghosts (no BC)	equilibrium solve in ghosts (no BC)
Radiation	in radiative shadow (large column)	unattenuated stellar spectrum
Type	isothermal–isochoric reservoir + valve	supersonic free outflow
Code	Apply_BC.f90: 32--54	Apply_BC.f90: 73--77

Table 1: Boundary conditions at a glance. Ghost cells: $N_g = 2$ per side; BCs act on ghosts only and reach the interior through the HLLC fluxes.

cold density, but seeding the Parker *density* itself tips the breathing base from self-limiting to self-amplifying and crashes the run.

7. Empirical base-BC test matrix (HD209458b)

To pin down how the lower-boundary choices above actually affect a converged wind, HD209458b was relaxed under a controlled combination of solver and base options (the runs live in EXHALE/HD209458b_test/). Each folder name encodes the model as <base>_<solver>_He{0,1}_met{0,CNO,FULL}[_flags]: **base** is dens ($n_0 = 10^{14} \text{ cm}^{-3}$ fixed; the dens12p5 run lowers it to $\log n_0 = 12.5$) or pres1ub (pressure-anchored at $1 \mu\text{bar}$, §2.5); **solver** is PLM (single-stage, stopped at a *du* plateau) or PWN (two-stage PLM→WENO3 then a Newton/JFNK finish); **S/V** are the Shapiro filter / mass-flux base velocity of §2.5; **He/met** are He 2^3S and the trace metals (C,N,O or the full C,N,O,Mg,Ca,Na,Fe set); **IC** is the cold hydrostatic or the Wind-AE warm start (§5.).

folder	BC	solver	S	V	He	met	IC	$\log \dot{M}$	v_{out}	neg.%	stop
<i>Pressure-anchored base ($1 \mu\text{bar}$, $n_0 \approx 5 \times 10^{12} \text{cm}^{-3}$)</i>											
pres1ub_PWN_He0_met0_cold	pres	PWN	—	—	—	—	cold	10.39	+8.59	0.4	Newton
pres1ub_PWN_He1_metCNO_cold	pres	PWN	—	—	✓	CNO	cold	9.53	+6.63	31	JFNK fail → stall
pres1ub_PWN_He1_metFULL_cold	pres	PWN	—	—	✓	FULL	cold	9.37	+6.29	36	JFNK fail → stall
pres1ub_PWN_He1_metCNO_cold_Vflux	pres	PWN	—	✓	✓	CNO	cold	9.53	+6.63	31	JFNK fail → stall
pres1ub_PWN_He1_metCNO_cold_Shap	pres	PWN	✓	—	✓	CNO	cold	—	—	—	running [†]
pres1ub_PLM_He0_met0_cold	pres	PLM	—	—	—	—	cold	10.40	+8.36	0.2	stall
<i>Density-anchored base ($n_0 = 10^{14} \text{cm}^{-3}$, except dens12p5: $\log n_0 = 12.5$)</i>											
dens_PWN_He0_met0_cold	dens	PWN	—	—	—	—	cold	10.50	+8.80	0.6	Newton
dens12p5_PWN_He0_met0_cold	dens	PWN	—	—	—	—	cold	10.38	+8.58	0.2	Newton
dens_PWN_He1_metCNO_cold	dens	PWN	—	—	✓	CNO	cold	9.61	+6.79	36	JFNK fail → stall
dens_PWN_He0_met0_windae_spherical	dens	PWN	—	—	—	—	windae	10.37	+5.77	13	breathing [‡]
dens_PLM_He0_metCNO_cold	dens	PLM	—	—	—	CNO	cold	9.72	+6.77	44*	stall
dens_PLM_He0_met0_cold_Vflux	dens	PLM	—	✓	—	—	cold	10.50	+8.58	1.0	stall
dens_PLM_He0_met0_cold_Shap	dens	PLM	✓	—	—	—	cold	—	−13.34	99.6	infall
dens_PLM_He0_met0_cold_Shap_Vflux	dens	PLM	✓	✓	—	—	cold	—	−13.32	99.6	infall
dens_PLM_He0_met0_windae_spherical	dens	PLM	—	—	—	—	windae	10.08	+3.68	31	du-stop
dens_PLM_He0_met0_windae_spherical_Shap_Vflux_Force	dens	PLM	✓	✓	—	—	windae+F	—	−20.03	86	infall

Table 2: HD209458b base-BC test matrix. $\dot{M}$ in g s^{-1} ; v_{out} in km s^{-1} ; neg.% is the percentage of cells with $v < 0$. “—” marks runs on an infall branch (wind $\dot{M}$ undefined). “JFNK fail → stall” = the Newton steady solver did not converge, so the run fell back to time-marching and stopped at a du plateau; those are not Newton-converged (carry the $\sim 2\text{--}5\%$ path-dependent $\dot{M}$ spread of the PLM-stall runs). The Newton rows reached a genuine solution (solver info=0). *The 44% are $\sim 2 \text{ms}^{-1}$ near-hydrostatic base noise. [†]The pressure + Shapiro run had not converged after $>0.8\text{M}$ steps when this table was written (Shapiro greatly slows / destabilizes convergence here). [‡]Wind-AE warm start: JFNK never triggered ($\|R\|$ stayed above the switch); the Wind-AE IC marched under WENO3 drifts to an unconverged breathing state ($du \sim 0.7$), mostly

What the matrix shows.

- **The Shapiro filter drives infall.** On the cold-IC density runs, every S-on case relaxes to a global infall ($v < 0$ in 99.6% of cells) while the matched S-off case is a clean outflow, independent of the velocity BC — the headline reason the filter is opt-in (§2.5). On the pressure + full-physics case the filter instead prevents convergence altogether (row †).
- **The base BC barely changes the converged wind — the lever is the base *density*.** With a Newton finish, density ($n_0 = 10^{14}$) and pressure ($1 \mu\text{bar}$) bases give nearly the same $\dot{M}$ (10.50 vs. 10.39). A *density*-anchored run at the same low density (dens12p5, $\log n_0 = 12.5$) reproduces the pressure result almost exactly (10.38 vs. 10.39; +8.58 vs. +8.59), confirming that pressure-anchoring acts only through the lower base density, not through the anchoring variable itself. The mass-flux velocity (V) is benign (it reproduces the valve result).
- **Physics lowers $\dot{M}$.** He 2^3S + C/N/O cooling drops $\log \dot{M}$ from ~ 10.4 to ~ 9.5 – 9.6 ; the full metal set (adding Mg/Ca/Na/Fe line cooling) lowers it further to 9.37. With full physics the JFNK solve fails for both bases and falls back to a stall.
- **The Wind-AE warm start does not help here.** It either du-stops at step 2 (returning $\approx$ the IC) or, with a Newton finish, never triggers JFNK and drifts to an unconverged breathing state. The cold hydrostatic IC, consistent with EXHALE's own base, is the reliable choice.